2ND SEMESTER LAB REPORT

TITLE : INTRODUCTION TO BASIC LAB EQUIPMENTS



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Submitted To:

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Lab Engineer

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EXPERIMENT NO 1

Lab Title: DIODE CHARACTERISTICS AND BASIC GATES

Objective: **TO learn basic equipments and their uses in ECA lab**

Student Name: Muhammad Waqas **240756**

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: _____ Signature: _____

LABORATORY EXPERIMENT NO. 1

INTRODUCTION TO BASIC LAB EQUIPMENTS

OBJECTIVE

1.To get familiarization with Digital Multimeter.

2. To get accustomed with Bread Board.

3. Introduction to DC Power Supply, Oscilloscope and Function Generator.

4. To know their usage, operation and basic safety requirements.

DISCUSSION:

An **oscilloscope** is a device used to visualize electrical signals. It displays how a voltage changes over time on a screen, with the **x-axis** showing time and the **y-axis** showing voltage. It helps analyze waveforms, measure frequency, and check circuit performance. Engineers and technicians use it to troubleshoot and design electronic systems.

How It Works

- When you connect an electrical signal to the oscilloscope, it converts the signal into a moving waveform.
- The waveform helps in understanding signal behavior, such as amplitude (voltage level**),** frequency, and shape.
- By adjusting the oscilloscope's settings, you can zoom in on small details or observe long-term trends.

Main Controls

- **Vertical Controls** – Adjust the height (voltage) of the waveform.
- **Horizontal Controls** – Adjust the time scale to see fast or slow changes.
- **Trigger Controls** – Stabilize a repeating signal so it doesn't move around the screen

---------(Lab Tasks)----------

Task1:)- Turn On the oscilloscope and adjust the necessary controls to establish a clear, bright, horizontal line across the centre of screen.

Turn on the oscilloscope and make sure the screen is displaying a clear, bright line. Adjust the intensity knob to control brightness and the focus knob to sharpen the line. Use the horizontal position and vertical position knobs to center the line on the screen. If the line is not steady, adjust the time base and trigger controls until you see a stable horizontal line across the middle. o stabilize the display, check the time base settings, which control how fast the signal moves across the screen, and adjust the trigger lev**el** to lock the signal in place. Once these settings are correctly adjusted, you will see a clear, bright, horizontal line across the center of the screen, indicating that the oscilloscope is properly set up and ready for measurements.

Harware Implementation:



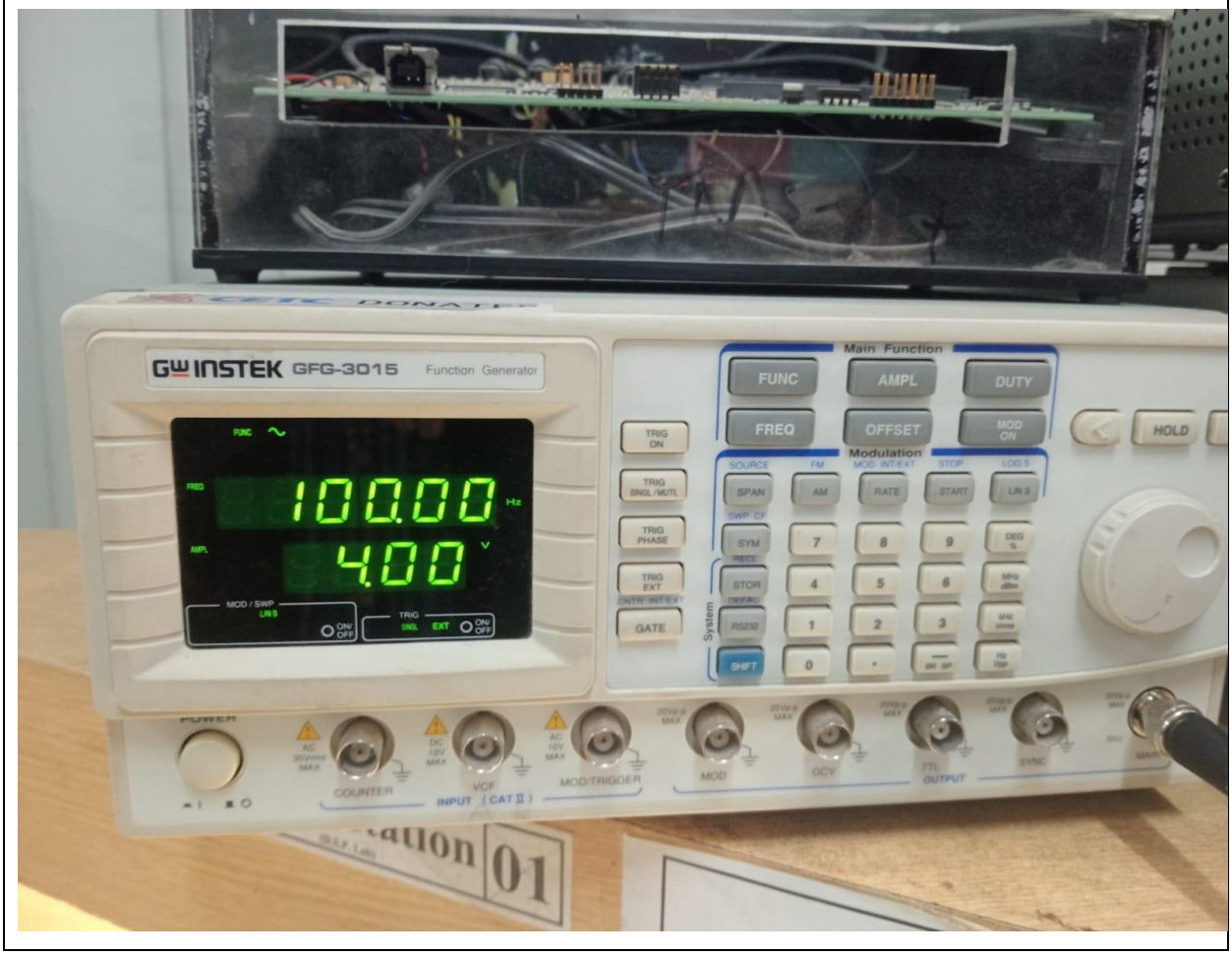
Task2:)- Turn Connect the function generator to one vertical channel of the oscilloscope and set the output of the generator to a 100Hz sinusoidal waveform

First, connect the function generator to the oscilloscope using a cable, making sure the ground wires are connected properly. Turn on the function generator and set it to produce a sine wave at 100 Hz (Hertz). Adjust the signal strength (amplitude) so it is clearly visible. On the oscilloscope, select the correct input channel and adjust the time and voltag**e** settings to display the waveform properly. If the wave is moving too much, adjust the **t**rigger settings to make it stable. After these steps, you should see a smooth 100 Hz sine wave on the oscilloscope screen.

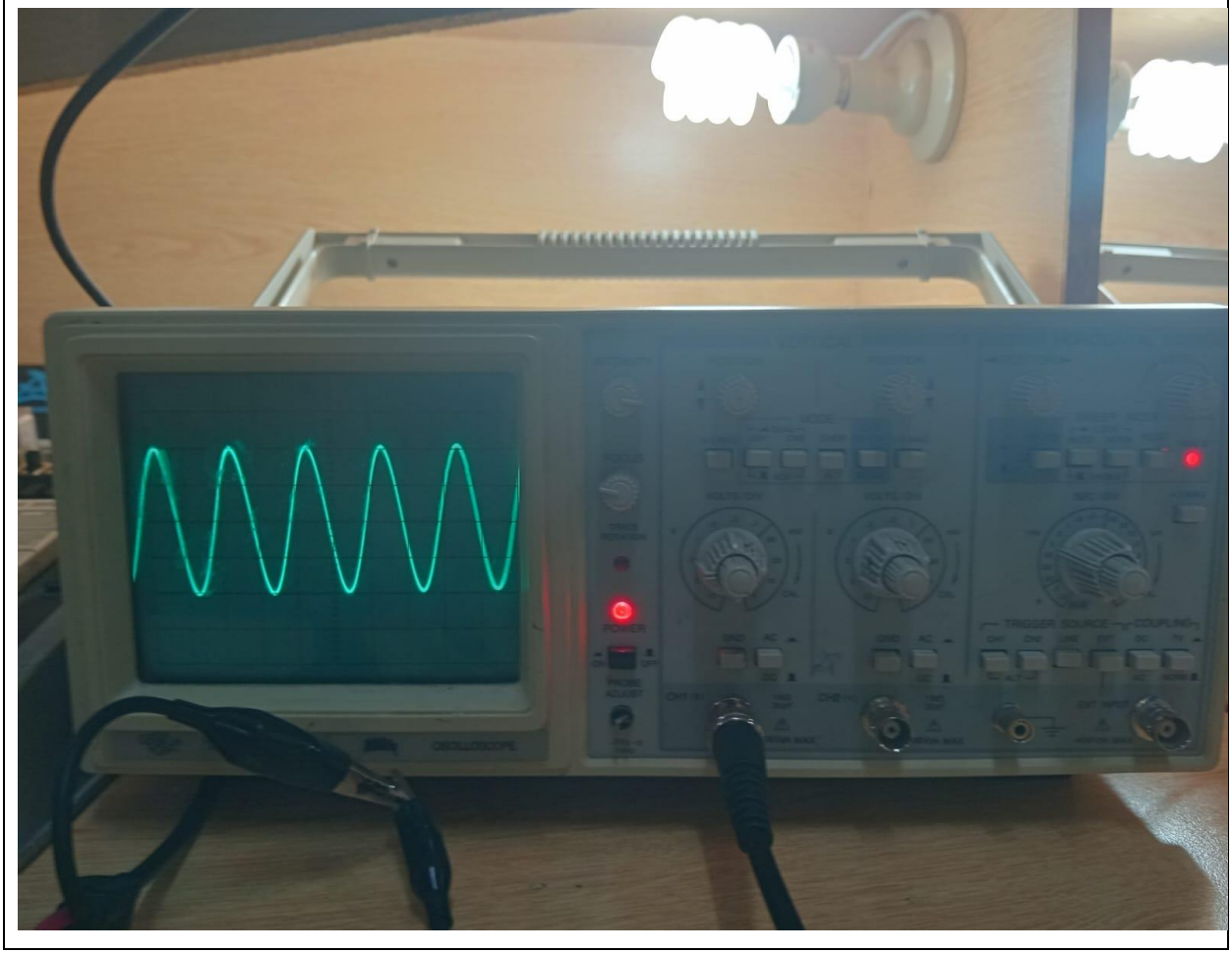
FUNCTION GENERATOR:

A **function generator** is an electronic device that produces different types of electrical waveforms, such as **sine waves, square waves, and triangular waves**. These waveforms are used for testing and troubleshooting circuits.

----------(FUNCTION GENERATOR)----------



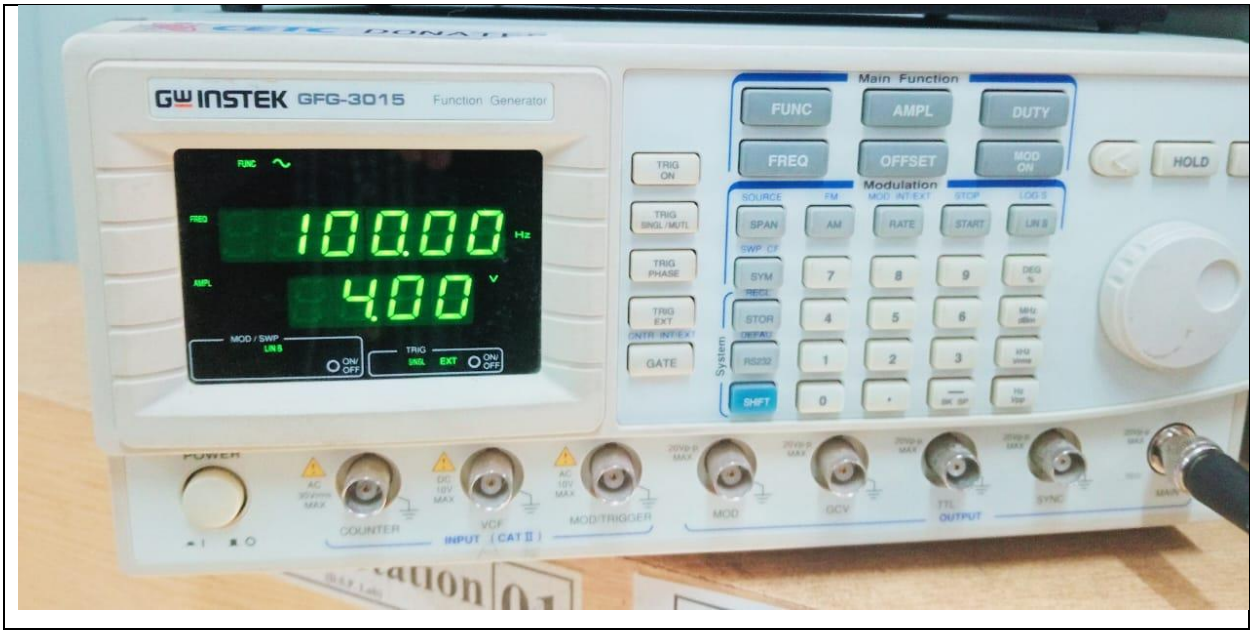
- **100Hz sinusoidal waveform**



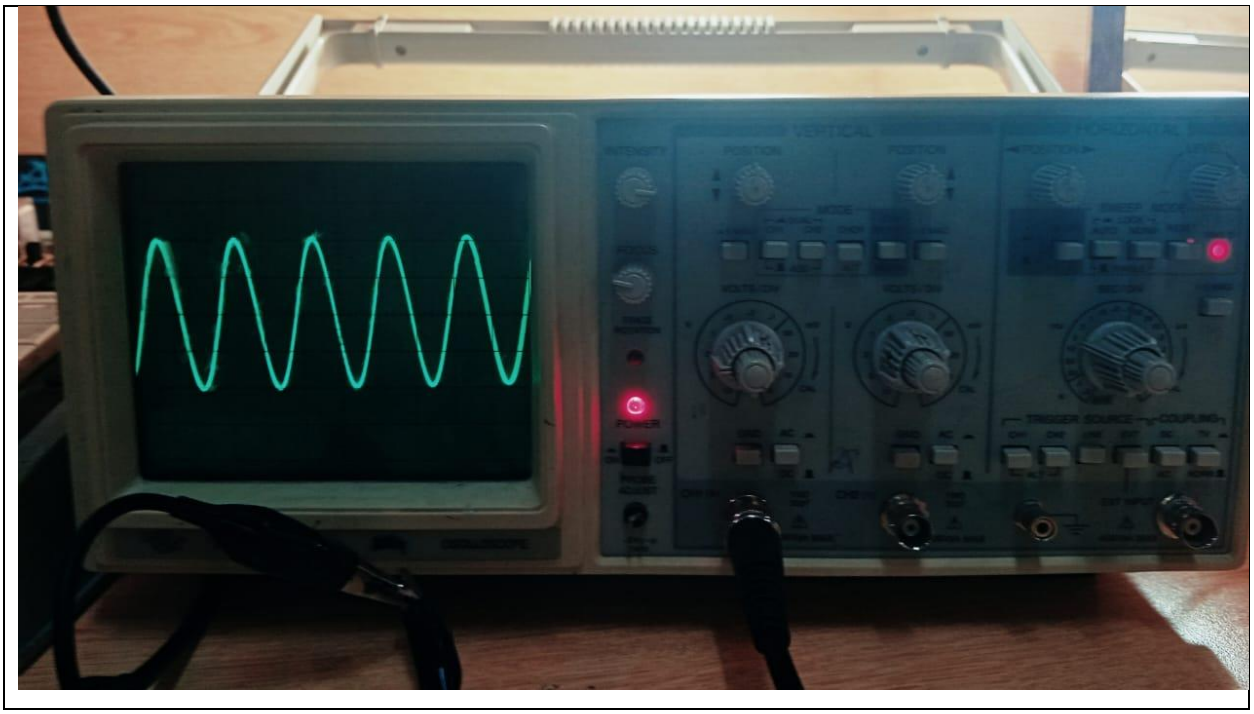
Task3:)- Set the vertical sensitivity of the scope to 1V/cm and adjust the amplitude control of the function generator to establish a 4V peak to peak sinusoidal waveform on screen.

To set up the oscilloscope for measuring a 4V peak-to-peak sine wave, first, adjust the vertical sensitivity (also called the volts per division setting) to 1V/cm. This means that each vertical division on the oscilloscope screen represents 1 volt. Now, turn on the function generator and set it to produce a sinusoidal waveform. Slowly adjust the amplitude control of the function generator while observing the waveform on the oscilloscope screen. Keep increasing or decreasing the amplitude until the waveform reaches 4 divisions from peak to peak (top to bottom), ensuring it measures exactly 4V peak-to-peak. If needed, fine-tune the oscilloscope's vertical position to keep the waveform centered on the screen. Once properly adjusted, you should see a stable 4V peak-to-peak sine wave, ready for further observation or measurement.

Function Generator with set values:



A sinusoidal wave has formed having 4 amplitude



Conclusions:

- We learn about oscilloscope and Its output display on screen.
- We learn it displays how a voltage changes over time on a screen, with the **x-axis** showing time and the **y-axis** showing voltage.
- We learn how oscilloscope helps analyze waveforms, measure frequency, and check circuit performance.
- Engineers and technicians use it to troubleshoot and design electronic systems.
- If something is wrong with an electrical circuit, you can often see it on the oscilloscope.
- Learn about different wave formation due different signals such as **sine waves, square waves, and triangular waves**.
- These waveforms are used for testing and troubleshooting circuits.